

Country-specific correlations across Europe between modelled atmospheric cadmium and lead deposition and concentrations in mosses.

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Previous analyses at the European scale have shown that cadmium and lead concentrations in mosses are primarily determined by the total deposition of these metals. Further analyses in the current study show that Spearman rank correlations between the concentration in mosses and the deposition modelled by the European Monitoring and Evaluation Programme (EMEP) are country and metal-specific. Significant positive correlations were found for about two thirds or more of the participating countries in 1990, 1995, 2000 and 2005 (except for Cd in 1990). Correlations were often not significant and sometimes negative in countries where mosses were only sampled in a relatively small number of EMEP grids. Correlations frequently improved when only data for EMEP grids with at least three moss sampling sites per grid were included. It was concluded that spatial patterns and temporal trends agree reasonably well between lead and cadmium concentrations in mosses and modelled atmospheric deposition. Copyright © 2012 Elsevier Ltd. All rights reserved.